LEF MODEL · CHAPTER 10

Chapter 10: The Chemical Fingerprint — the Iron Rings, the Thorium Riddle in the Contact Zone and the Five Clocks

Based on Paper 05 — The Maria (Parts J–O and Appendix A: Studies 9, 10, 11, 13 and the summary of the 16 studies)

01 · THE MYSTERY

The mystery: why is the Moon's radioactivity concentrated in one circle — and why are the mare floors poor in it?

One of the most surprising discoveries of the Apollo missions and the Lunar Prospector satellite is the extreme chemical inequality of the Moon's face: the radioactive element **thorium** — together with potassium, rare-earth elements and phosphorus (a material called **KREEP** in the literature) — is not spread evenly over the Moon, but is concentrated almost entirely in one patch in the west of the near side.

The accepted literature interprets this patch (the "Procellarum KREEP Terrane") as the last remnant of a global "magma ocean" that crystallised on the Moon, and holds that the radioactive heat of the KREEP is what melted the rocks of the near side and created the basaltic maria. But as soon as the data are examined at high resolution, **three sharp contradictions** of this explanation appear:

1. **The exact geometric centre:** why does the thorium patch sit precisely around the point 24.1°N, 335.2°E — exactly the point fixed in advance in

Paper 3 as **the centre of the last contact disc** between the Moon and the Earth?

2. **The paradox inside the contact zone:** if KREEP heat is what broke out of the maria and brought the thorium with it, the mare floors should be the richest in thorium. In fact the exact opposite happens: **the mare floors inside the contact zone are the poorest ground in thorium in that region!**

3. **The absence of foreign mass in the basins:** if the great mare basins (such as Imbrium and Serenitatis) were dug by giant asteroids from outer space, their melt rocks should contain the fingerprint of the iron-loving metals (**siderophile elements**) of those foreign bodies — but they do not!

02 · AN IMAGE

An image: freezing seawater (the thorium retreats with the brine) and a firing range that keeps its distances

To see how these three chemical mysteries are solved in one stroke, imagine two simple experiments from everyday life:

- **Seawater freezing in a tank connected to a drain pipe (the thorium riddle):** when salty seawater begins to freeze from the top, the ice crystals that form are almost fresh water — and all the salt is pushed into the liquid brine left below. If this brine sits in a closed plain with no drainage (like **the basalt plains** that melted in place), the salt stays trapped in the rock when everything freezes. But if the brine sits above **a pipe open downward** (like the floor of a **mare**, connected to the feeding channel into the depths!), the heavy, thorium-rich brine sinks and drains back down the channel into the depths — leaving the mare floor poor in thorium compared with the plains and highlands around it!
- **A firing range of light magma early and far, and heavy magma late and near (the iron rings):** at the start of a feeding opening's activity, the magma is rich in volatile gases and poor in iron, so it is very light and fluid — and is fired out to hundreds of kilometres (up to 600 km). As the gases escape, the magma becomes heavier, more viscous and richer in iron (FeO) and titanium (TiO_2),

and lands very close to the rim of the opening — creating **a graded iron halo** around every mare!

03 · SCIENCE AND MODEL

Science and model: the iron and titanium rings, 95% of the thorium in the contact disc, and the absence of foreign mass

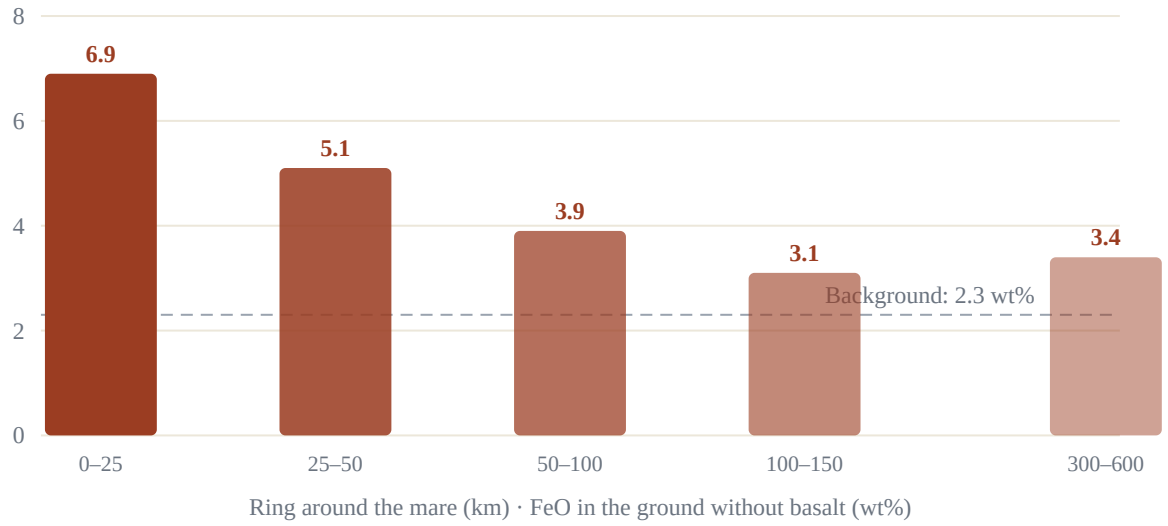
Part J of Paper 05 examines the satellite chemistry maps of Kaguya (iron and titanium at 240 m resolution) and of Lunar Prospector (thorium at 60 km resolution) and presents four decisive findings:

1. **The iron and titanium halo around the maria reaches 600 km (Study 9, §J.1–J.2):** around six maria (Smythii, Imbrium, Serenitatis, Crisium, Moscoviense and Nectaris), after removing all basalt pixels, the iron content of the ground drops gradually from **6.9 wt%** in the 0–25 km ring, to 5.1 at 25–50 km, to 3.9 at 50–100 km, and to **3.1 wt%** at 100–150 km, and stays high above background levels (2.3 wt%) even at **300–600 km (3.4 wt%)** — ten times the range of sideways mixing by later craters! By contrast, around **Oceanus Procellarum** — a basalt plain with no feeding opening that merely spilled over its edges — the iron drops from 10.9 wt% to 2.4 wt% (the level of the far highlands) at 300–600 km.

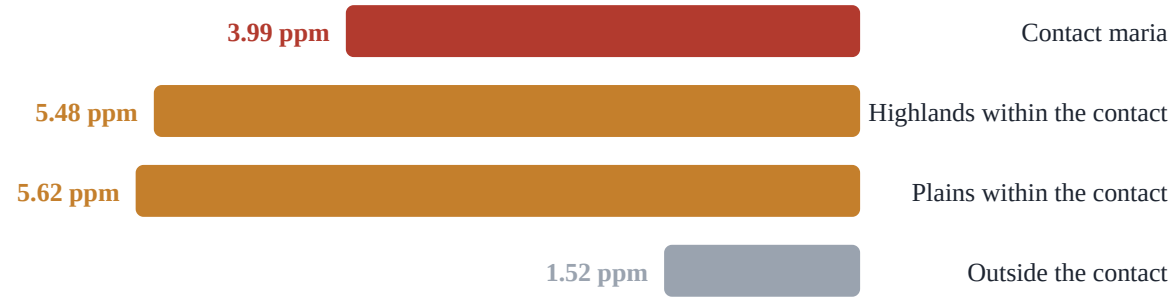
2. **95% of the Moon's thorium excess is trapped in the contact disc (Study 10, §J.3):** the contact disc (radius 1,500 km around 24.1°N, 335.2°E) covers only **17.5%** of the Moon's surface, yet it holds **95.0% of all the thorium excess above 3 ppm** (median 5.39 ppm inside the contact against 1.52 ppm outside it). When a disc of the same size is moved over **16,200 possible centre points** across the whole Moon, the contact centre fixed in advance in Paper 3 ranks **third out of 16,200** (only 47 km from the theoretical maximum point, which gathers 95.14% against 94.99%)!

3. **Inside the contact zone — the maria are poor in thorium while the plains and highlands are identical (Study 13, §J.4):** inside the contact disc, the basalt plains (5.62 ppm) and the highlands (5.48 ppm) carry almost exactly

the same thorium concentration! By contrast, the four maria within the contact (Imbrium, Serenitatis, Copernicus H and Sinus Aestuum) are **30%** poorer in thorium than the plains (only 3.99 ppm, and each of the four maria is poorer than the ring around it by 0.58–1.39 ppm), while their titanium content stays almost unchanged (−5%, ratio $\mathrm{Th}/\mathrm{TiO_2} = 0.74$) — because the titanium is locked in ilmenite crystals and stays, while the last thorium-rich liquid drained down the feeding channels!



The iron halo around six maria (Study 9): the iron falls with distance, but stays above the background out to 600 km.



Thorium inside and outside the contact disc (Studies 10 and 13): the plains and highlands are identical, and the maria are 30% poorer — the thorium-rich liquid drained down the channels.

The foreign-mass test in the basins (Study 11, §J.7)

Had the great mare basins been dug by bodies from outer space at a speed of 18 km/s, they would have required an impacting mass of about **1×10^{19} kg** (and for Imbrium alone 3×10^{18} kg by scaling, or 2×10^{19} kg for a body 250 km across, which would have left 6×10^{17} to 4×10^{18} kg of foreign material). In fact, the ratios of siderophile elements in the melt rocks of Imbrium **do not match any known chondrite group**, and most of them already existed in the crust before Imbrium and Serenitatis — because the basins are **terrestrial feeding openings**, not impacts of foreign bodies!

04 · THE MECHANISM IN WORDS

The mechanism in words: the five clocks of every opening — and decoding the ages of Apollo and Chang'e

Part K of Paper 05 resolves one of the greatest misunderstandings in lunar research: **why are the basalt rocks brought back by space missions dated to 3.1–3.9 billion years (Apollo), 2.8 billion years (Chang'e-6 on the far side) and even 2.0 billion years (Chang'e-5 in Oceanus Procellarum)?** Does this mean that the openings formed billions of years after the Moon left the Earth?

The answer is **no** — because every feeding opening carries **five separate clocks** that must not be mixed (like a well that is dug, opened, carries water, is sealed, and only at the end does the last puddle at its top dry up):

1. **The time of creation of the structure ($t_{\text{structure}}$):** the moment the opening and the dome beneath it were created during the building stage (during the 600 million years before the detachment at 3.873 Ga).
2. **The time of exposure (t_{exp}):** the moment the edge of the contact retreated above the opening and exposed it to space (generation I at once, generation II during the rounding, generation III toward the detachment).

3. **The time of eruption (t_{vent}):** the period during which the opening fired material outward, creating the craters and the iron rings around it.

4. **The time of closure (t_{shutdown}):** the moment the flux fell below the critical threshold (or the pressure stopped at the moment of detachment) and the opening ceased to be active ($t_{\text{shutdown}} \geq t_{\text{exp}}$).

5. **The age of the surface (t_{surface}): the age of final cooling** of the last drop of magma that froze on the surface. A radiometric clock begins to tick only from the moment the rock cools below its closure temperature and traps the radioactive daughter atoms!

So all the basalt rocks of Apollo, Chang'e-5 and Chang'e-6 measure t_{surface} **— the time of final freezing of the hot liquid** (kept hot for a long time by the Moon's internal heat or by the concentration of thorium in the contact zone), and not the time the opening opened ($t_{\text{structure}}$)!

05 · THE KEY EQUATIONS

Academic explanation and the key equation

The chemical ratio that distinguishes the mare floors from the basalt plains inside the contact zone, and the distribution of thorium by quarters of the Moon, are summarised in the following expressions:

$$\frac{(\mathrm{Th/TiO_2})_{\text{mare floor}}}{(\mathrm{Th/TiO_2})_{\text{plain}}} = 0.74 \quad (0.60\text{–}0.88) \quad \text{within the contact}$$

$$\mathrm{Th}_{\text{contact}} = 5.39 \text{ ppm} \quad \text{against} \quad \mathrm{Th}_{\text{outside}} = 1.52 \text{ ppm}$$

By longitude (§J.5), the median thorium concentration falls gradually from the near quarter facing the Earth (**5.12 ppm**), to the leading half around 90°W in which the contact centre lies (3.27 ppm), to the far side (2.20 ppm) and to the trailing half around 90°E (2.01 ppm) — 1.5 times higher over the whole near side than on the far side (2.56 against 1.76 ppm).

The signature in reality: one unified picture across 15 studies and 34 catalogued openings

Parts L–O of Paper 05 unite the five independent lines of evidence into a single complete feeding system:

- **The balance of the studies (13 green and 2 amber out of 15 studies, §M.2):** all the main studies were fully verified — the match of 33/34 maria to the gravity domes, the 44 buried domes, the crustal step (17 km), the sinking of the centre (10/12), the uniform level (16/16), the clean floors (12 times and 83 times), the iron halos (600 km), the thorium centre (3/16,200), the absence of foreign mass, and the wear of rims on the contact plains. The two studies that received an "amber" light (partial confirmation) were reported with full transparency: in Study 13 the ratio $\mathrm{Th}/\mathrm{TiO}_2$ also fell beyond the contact because of a relative richness of titanium in the floors, and in Study 16 the firing collar is significant around the far-side maria as a whole (p = 0.009) and around the buried ones (p = 0.001), but does not separate from the control around the 10 open far-side maria alone.
- **Five decisive questions that separate the model from the literature (§N.2):** the literature cannot explain how asteroids created 44 gravity domes without basins above them, why a mare floor freezes at a single level and sinks only above a strong dome, why the iron halo reaches 600 km, why the thorium is identical on the plains and highlands of the contact and low precisely in the maria, and where 10^{19} kg of external metal disappeared to.

The lab box: the chemical table, a sample of the main openings and a summary of the 8 conclusions (Paper 05, Parts J–O and Appendix A)

Study / chemical measure	Region / comparison group	Measured finding	Meaning in the LEF model

Iron (FeO) and titanium (TiO_2) rings (Study 9)	**6 maria combined (0–25 to 300–600 km)**	**FeO: $6.9 \rightarrow 3.1$ wt% (150 km), stays at 3.4 at 600 km (control 2.3)**	**The maria fired light material far and heavy material near; at Procellarum it falls to 2.4 (local spill)**
Thorium concentration in the contact disc (Study 10)	**The contact disc (17.5% of the Moon's surface) against 16,200 centres**	**95.0% of the thorium excess (>3 ppm); median 5.39 against 1.52 ppm**	**The contact centre ranks 3 out of 16,200 (47 km from the peak); the mark of the last hold**
Division of thorium within the contact (Study 13)	The 4 maria against the plains and highlands within the contact	Maria: **3.99 ppm** (−30%); plains: **5.62**; highlands: **5.48**	The last thorium-rich liquid drained back down the mare channels
The foreign mass of the basins (Study 11)	All mare basins (Imbrium first) at 18 km/s	About 10^{19} kg required; in fact **no match to any chondrite**	There were no foreign impactors — the basins are terrestrial feeding openings

# in the catalogue	Name of the opening (Appendix A)	Coordinates and distance from sub-Earth	Zone and generation of exposure	Feed dome (GRAIL) and diameter	Cover today
#1	**Serenitatis**	**26°N, 18°E (31°)**	**Contact (generation III)**	**M660_008 — 610 km**	**Open (beneath a mare)**
#34	**Imbrium**	**38°N, 341°E (42°)**	**Contact (generation III)**	**M660_015 — 671 km**	**Open (beneath a mare)**
#35–36	Copernicus H / Sinus Aestuum	6°N, 342°E / 10°N, 352°E	Contact (generation III)	M660_042 (170) / M660_020 (235 km)	Open (beneath a mare)
#3–4, #6, #33	Nectaris, Crisium, Smythii, Humorum	Distance 36°–86°	Hill (generation II)	375–518 km	Open (beneath a mare)
#32	Schiller–Zucchius	57°S, 316°E (67°)	Hill (generation II)	M660_014 — **909 km** (the largest)	Buried (beneath the highland)

#10, #26	Moscoviense / Orientale	140° / 95° (far side)	Beyond the hill (generation I)	M660_002 (313) / M660_001 (435 km)	Open (exceptions on the far side)
#7, #9, #13, #18–19, #23–24, #37	Milne, Mendeleev, Freundlich–Sharonov, Korolev, Hertzsprung, Schrödinger and more	100°–162° (far side)	Beyond the hill (generation I)	106–346 km (of the 36 buried domes on the far side)	Buried beneath the far-side layers
#12, #20, #29	Ingenii, Apollo, Bailly	82°–146° (above SPA)	Beyond the hill (generation I)	Their dome was cooked and erased by the heat of the South Pole–Aitken basin	Above the SPA basin

07 · NEXT

The door ahead

In Chapters 8–10 we completed the investigation of the Moon's maria: it turns out that they are not late impact wounds, but **the ancient feeding openings** through which the Moon was built above 73 gravity domes — openings that closed in the order of the contact's retreat, froze as lake levels that sank at their centres, painted their surroundings with iron rings out to 600 km, and left in the contact disc the thorium mark of the last hold on the Earth.

But if the maria were the barrels from which material was fired outward, **what happened to this material after it was fired into space — and how did it come back and strike the Moon to dig the hundreds of thousands of craters on its face?** In the next chapter — **Chapter 11: The Craters of the Moon — 99% Returning Material, Not Foreign Asteroids** (from Paper 06 — *The Craters*) — we will decode the story of the craters!

Chapter 11

The Craters of the Moon — 99% Returning Material, Not Foreign Asteroids →

← Chapter 9 · Prologue

© 2026 Zohar Duran · The LEF Model · lef-model.com · ORCID 0009-0005-3194-9387